

CMCMIS_SIMPLIFIED

DUMP90TABLES SCHEMA INTELLIGENCE REPORT

Senior Engineering Extraction | Database Evidence | Phase 6+ Backbone

LOCKED PURPOSE

This document converts the uploaded dump90tables.zip / dump90tables.sql into a build-ready engineering map: table inventory, columns, indexes, foreign keys, module ownership, risk audit, and Phase 6 Slice 2 implementation direction. It is a parsed-dump report, not a live DB introspection result.

Document Control

Field	Value
Source archive	dump90tables.zip
Source SQL	dump90tables.sql
Source SHA256	dd57e353524636a960f04c0fcd658fb8428ecd6e88e653e52a399b81381ffc6a
Extracted tables	90
Extracted columns	966
Indexes parsed	172
Foreign keys parsed	84
INSERT tuple rows counted	398,329
Status	LOCKED - Parsed evidence bundle generated

1. Executive Lock Snapshot

ONE-LINE RESULT

The uploaded dump contains 90 tables, 966 columns, 172 indexes, 84 foreign keys, and 398,329 inserted data rows. It is now mapped into CSV, SQL, JSON, Markdown, and this DOCX report.

Database Evidence Metrics

Metric	Value	Engineering meaning
Total tables parsed	90	Full dump90 table set confirmed.
Active/non-legacy tables	75	Available for current/future modules after validation.
legacy* tables	15	Preserved isolation layer; read-only unless bridge approved.
Columns parsed	966	Column catalog generated for repository mapping.
Indexes parsed	172	Performance and uniqueness map extracted.
Foreign keys parsed	84	Relationship graph extracted for safe joins and transactions.
Rows counted from dump	398,329	Approximate historical data mass represented in INSERT statements.
Tables without PK	19	High-priority risk bucket before write APIs.

2. Delivered File Bundle

Folder/File	Purpose
csv/01_table_inventory.csv	Main table-by-table inventory with module, rows, columns, PK/FK/index counts, phase mapping.
csv/02_column_catalog.csv	Every column with raw data type, nullability, defaults, enum values, PK/FK flags, role guess.
csv/03_index_catalog.csv	Primary, unique, secondary, and fulltext index map.
csv/04_foreign_key_catalog.csv	FK constraints, source/target tables and columns, delete/update actions.
csv/05_relationship_edges.csv	Table-level graph edges for ERD/relationship visualization.
csv/06_module_summary.csv	Aggregated table/column/row/FK counts by CMCMS module.
csv/07_table_risk_audit.csv	Risk observations: no PK, possible missing FK, date-as-text, charset legacy, security field issues.
csv/08_phase_mapping.csv	Table-to-phase and recommended engineering action map.
csv/09_row_counts_from_dump.csv	Data mass per table counted from INSERT tuples.
csv/10_enums_and_status_fields.csv	Status/state/ENUM field catalog for state-machine work.
sql/00_schema_only_extracted_from_dump90tables.sql	Schema-only DDL extracted from the dump; all INSERT data removed.
sql/01_senior_review_queries.sql	Post-import validation queries for table count, PKs, FKs, row counts.
sql/02_phase6_slice2_db_action_plan.sql	Phase 6 Slice 2 DB pre-flight and transaction doctrine SQL notes.
ddl_per_table/*.sql	One DDL file per table for focused review and module-level repository mapping.
json/schema_catalog_dump90tables.json	Machine-readable schema catalog for future tooling or AI-assisted code generation.
markdown/*.md mmd/.dot	Human-readable reports and relationship graph source files.

3. Module Summary - What the Dump Is Really Saying

Module	Tables	Columns	Rows	FKs	Key tables
Audit / Security	3	23	20	1	audit_log audit_log_changes export_audit
Auth & RBAC	7	51	387	7	login_audit permissions refresh_tokens roles role_permissions users user_roles
Equipment / Instruments	9	141	10,354	17	cmms_eqipinst_identification cmms_eqip_detail_spec cmms_eqip_mst cmms_eqip_mst_hist cmms_eqip_tec_spec cmms_ins_accuracy_info cmms_inv_eqip_dtl cmms_schedule_eqip_dtl equipment_status_history
Job Cards	18	254	302,946	21	cmms_fault_mst cmms_jobcard_attendedby_dtl cmms_jobcard_awaitinginfo cmms_jobcard_cal_adjustments_dtl cmms_jobcard_cal_dtl cmms_jobcard_cal_observations cmms_jobcard_contract_warranty_dtl cmms_jobcard_eq_used cmms_jobcard_faulty_category...
Job Requests	5	79	48,911	10	cmms_jobrequest_item_dtl cmms_jobrequest_mst cmms_jobrequest_project_dtl job_request_accessories job_request_status_history
Legacy / Isolated	15	91	9,328	6	_legacy_accessright_mst _legacy_cal_jobcard_feedback_spec _legacy_cf001 _legacy_cf002 _legacy_cf003 _legacy_cf004 _legacy_chklistvendor _legacy_jobcard_insp_maint_dtl _legacy_module_mst _legacy_parameter_master_bkp _legacy_parameter_mast...
Master Data /	10	78	19,886	3	cmms_checklist_hist cmms_checklist_mst cmms_checklist_tasks cmms_checklist_tasks_hist cmms_device_spares_mst cmms_lineitem_mst cmms_parameter_master cmms_product_mst cmms_proj_mst cmms_task_mst

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Organ ization / Emplo yee	7	83	4,2 05	6	cmms_designation_mst cmms_desig_hist cmms_division_hist cmms_emp_mst cmms_section_mst departments sections
Other / Cross - cuttin g	4	20	898	0	cmms_documentno_mst schema_migrations sysdiagrams tmpcaldtl
Procu remen t / AMC / Invent ory	8	108	689	9	cmms_amc_mst cmms_inv_mst cmms_po_item_dtl cmms_po_mst cmms_po_payment_dtl cmms_po_receive_dtl cmms_pur_dtl cmms_pur_mst
Sched uling / PM- CAL	1	8	6	1	cmms_schedule_mst
Vendo r / Manuf acture r	3	30	699	3	cmms_cont_mst cmms_supplier_mfr cmms_vendor_doc_spec

SENIOR INTERPRETATION

The data mass is concentrated in Job Cards and Job Requests. That means Phase 6 is not a small feature layer; it is the operational history center of CMC MIS. Any approve/assign/close logic must be transactional, audited, and index-aware.

4. Top Data-Mass Tables

Largest tables by INSERT tuple count

Table	Module	Rows	Columns	PK
cmms_jobcard_cal_observations	Job Cards	77,171	4	JobcardNumber;TaskID
cmms_jobcard_eq_used	Job Cards	38,316	4	JEU_JobCardNo;JEU_EquipType;JEU_EquipId
cmms_jobcard_attendedby_dtl	Job Cards	27,890	4	JMA_SECTIONJOBNO;JMA_USERID;JMA_ISAWAITING
cmms_jobcard_status_hist	Job Cards	22,214	5	MISSING
cmms_jobcard_mst_history	Job Cards	22,143	37	HistoryId
cmms_jobrequest_mst	Job Requests	21,490	52	JR_JOBREQUESTNO
cmms_jobrequest_project_dtl	Job Requests	19,624	2	JR_JOBREQUESTNO;JR_PROJECTID
cmms_jobcard_mst	Job Cards	19,432	38	JM_SectionJobNo
cmms_jobcard_request_info	Job Cards	19,432	27	JRI_JobCardNo
cmms_jobcard_contract_warranty_dtl	Job Cards	17,225	17	CWD_JobCardNo
cmms_jobcard_request_item_dtl	Job Cards	11,064	10	JR_SECTIONJOBNO;JR_ITEM_ID
cmms_jobcard_cal_dtl	Job Cards	9,065	21	JCD_JobCardNo
cmms_jobcard_faulty_category	Job Cards	8,605	3	JobcardNumber;FaultyType;FaultyCategory
cmms_checklist_tasks_hist	Master Data / Lookup	8,450	4	MISSING
cmms_jobcard_faulty_section	Job Cards	8,131	3	JobcardNumber;FaultyType;FaultySection
cmms_jobcard_repair_info	Job Cards	8,118	18	JobcardNumber
cmms_jobrequest_item_dtl	Job Requests	7,786	11	JR_JOBREQUESTNO;JR_ITEM_ID
cmms_checklist_tasks	Master Data / Lookup	7,536	2	CLTSK_ID;CLTSK_TASKID
cmms_jobcard_awaitinginfo	Job Cards	7,261	19	JobcardNumber
cmms_eqip_mst	Equipment / Instruments	5,705	52	EQM_TYPE;EQM_ID
cmms_division_hist	Organization / Employee	3,676	7	MISSING
legacy_cf004	Legacy / Isolated	3,449	2	MISSING
legacy_accessright_mst	Legacy / Isolated	3,221	11	ACC_MOD_ID;ACC_ROLE
cmms_jobcard_spares equip	Job Cards	2,804	9	JobcardNumber;Sr_No
cmms_eqipinst_identification	Equipment / Instruments	2,286	10	EMD_EQIP_TYPE;EQM_ID;EII_ID

5. CMC MIS Phase Mapping

LOCKED IMPLEMENTATION VIEW

Do not treat all 90 tables equally. The next build must focus on Job Requests, Job Cards, Equipment references, Auth/RBAC, and Audit. Scheduling, Procurement, AMC, Reports, and legacy bridges remain future/read-only unless specifically activated.

Use category	Tables	Rows	Examples
ACTIVE_CORE_SECURITY	10	407	audit_log, audit_log_changes, export_audit, login_audit, permissions, refresh_tokens, roles, role_permissions

ACTIVE_MVP_CORE	32	362,211	cmms_eqipinst_identification, cmms_eqip_detail_spec, cmms_eqip_mst, cmms_eqip_mst_hist, cmms_eqip_tec_spec, cmms_fault_mst, cmms_ins_accuracy_info, cmms_inv_eqip_dtl
ACTIVE_SUPPORT_MASTER	20	24,790	cmms_checklist_hist, cmms_checklist_mst, cmms_checklist_tasks, cmms_checklist_tasks_hist, cmms_cont_mst, cmms_designation_mst, cmms_desig_hist, cmms_device_spares_mst
FUTURE_PHASE_CORE	9	695	cmms_amc_mst, cmms_inv_mst, cmms_po_item_dtl, cmms_po_mst, cmms_po_payment_dtl, cmms_po_receive_dtl, cmms_pur_dtl, cmms_pur_mst
LEGACY_ISOLATED	15	9,328	_legacy_accessright_mst, _legacy_cal_jobcard_feedback_spec, _legacy_cf001, _legacy_cf002, _legacy_cf003, _legacy_cf004, _legacy_chklistvendor, _legacy_jobcard_insp_maint_dtl
REVIEW_REQUIRED	4	898	cmms_documentno_mst, schema_migrations, sysdiagrams, tmpcaldtl

6. Phase 6 Slice 2 Core Tables

Job Requests

Table	Rows	PK	FKs	Action
cmms_jobrequest_item_dtl	7,786	JR_JOBREQUESTNO;JR_ITEM_ID	1	Target for Slice 2: detail/approve/reject/assign transitions.
cmms_jobrequest_mst	21,490	JR_JOBREQUESTNO	4	Target for Slice 2: detail/approve/reject/assign transitions.
cmms_jobrequest_project_dtl	19,624	JR_JOBREQUESTNO;JR_PROJECTID	2	Target for Slice 2: detail/approve/reject/assign transitions.
job_request_accessories	3	acc_id	1	Target for Slice 2: detail/approve/reject/assign transitions.
job_request_status_history	8	history_id	2	Target for Slice 2: detail/approve/reject/assign transitions.

Job Cards

Table	Rows	PK	FKs	Action
cmms_fault_mst	30	FM_FAULT_ID;FM_TYPE	0	Target for Slice 2/3: auto-create, execution, verify- close, PDF.
cmms_jobcard_attendedby_dtl	27,890	JMA_SECTIONJOBNO;JMA_USERID;JMA_ISAWAITING	2	Target for Slice 2/3: auto-create, execution, verify- close, PDF.
cmms_jobcard_awaitinginfo	7,261	JobcardNumber	1	Target for Slice 2/3: auto-create, execution, verify- close, PDF.
cmms_jobcard_cal_adjustments_dtl	1,831	JCAD_JobCardNo;JCAD_Parameter;JCAD_TestValue	1	Target for Slice 2/3: auto-create, execution, verify- close, PDF.
cmms_jobcard_cal_dtl	9,065	JCD_JobCardNo	1	Target for Slice 2/3: auto-create, execution, verify- close, PDF.
cmms_jobcard_cal_observations	77,171	JobcardNumber;TaskID	2	Target for Slice 2/3: auto-create,

				execution, verify-close, PDF.
cmms_jobcard_contract_warranty_dtl	17,225	CWD_JobCardNo	1	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_eq_used	38,316	JEU_JobCardNo;JEU_EquipType;JEU_EquipId	2	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_faulty_category	8,605	JobcardNumber;FaultyType;FaultyCategory	2	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_faulty_section	8,131	JobcardNumber;FaultyType;FaultySection	2	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_inspection_info	2,214	JobcardNumber	1	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_mst	19,432	JM_SectionJobNo	1	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_mst_history	22,143	HistoryId	0	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_repair_info	8,118	JobcardNumber	1	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_request_info	19,432	JRI_JobCardNo	1	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_request_item_dtl	11,064	JR_SECTIONJOBNO;JR_ITEM_ID	1	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_spare_parts	2,804	JobcardNumber;Sr_No	1	Target for Slice 2/3: auto-create, execution, verify-close, PDF.
cmms_jobcard_status_hist	22,214	MISSING	1	Target for Slice 2/3: auto-create, execution, verify-close, PDF.

Equipment / Instruments

Table	Rows	PK	FKs	Action
cmms_eqipinst_identification	2,286	EMD_EQIP_TYPE;EQM_ID;EII_ID	1	Use as asset source of truth; only controlled back-write from Job Card close.
cmms_eqip_detail_spec	3	EDS_EQIP_TYPE;EDS_EQIP_ID;EDS_FILENAME	1	Use as asset source of truth; only controlled back-write from Job Card close.
cmms_eqip_mst	5,705	EQM_TYPE;EQM_ID	6	Use as asset source of truth; only controlled back-write from Job Card close.
cmms_eqip_mst_hist	519	EQM_HIST_ID	0	Use as asset source of truth; only controlled back-write from Job Card close.
cmms_eqip_tec_spec	0	EMD_EQIP_TYPE;EQM_ID;EMD_SPEC_NAME	1	Use as asset source of truth; only controlled back-write from Job Card close.
cmms_ins_accuracy_info	1,501	EIA_INS_ID;EIA_TYPE;EIA_ACCURACY	1	Use as asset source of truth; only controlled back-write from Job Card close.
cmms_inv Equip Dtl	24	IVD_PARTNO;IVD_EQM_ID;IND_EQM_TYPE	2	Use as asset

				source of truth; only controlled back-write from Job Card close.
cmms_schedule Equip Dtl	316	SC_PLAN_ID;SC_TYPE;SC_EQM_ID;SC_EQM_TYPE;SC_SCHEDULE_DATE	2	Use as asset source of truth; only controlled back-write from Job Card close.
equipment_status_history	0	history_id	3	Use as asset source of truth; only controlled back-write from Job Card close.

Auth & RBAC

Table	Rows	PK	FKs	Action
login_audit	57	audit_id	0	Use only through auth/repository layer; permission checks by permission code, not role name.
permissions	40	permission_id	0	Use only through auth/repository layer; permission checks by permission code, not role name.
refresh_tokens	43	token_id	1	Use only through auth/repository layer; permission checks by permission code, not role name.
roles	5	role_id	0	Use only through auth/repository layer; permission checks by permission code, not role name.
role_permissions	122	role_id;permission_id	2	Use only through auth/repository layer; permission checks by permission code, not role name.
users	60	user_id	2	Use only through auth/repository layer; permission checks by permission code, not role name.
user_roles	60	user_id	2	Use only through auth/repository layer; permission checks by permission code, not role name.

Audit / Security

Table	Rows	PK	FKs	Action
audit_log	20	audit_id	0	Write in same transaction for state changes; never optional.
audit_log_changes	0	change_id	1	Write in same transaction for state changes; never optional.
export_audit	0	export_id	0	Write in same transaction for state changes; never optional.

Organization / Employee

Table	Rows	PK	FKs	Action
cmms_designation_mst	40	DG_ID	0	Use for user display, sections/divisions, engineer assignment dropdowns.
cmms_desig_hist	132	EMM_ID;EMM_DESIGNATION;EMM_DATE	1	Use for user display, sections/divisions, engineer assignment dropdowns.
cmms_division_hist	3,676	MISSING	1	Use for user display, sections/divisions, engineer assignment dropdowns.
cmms_emp_mst	60	EMM_ID	2	Use for user display, sections/divisions, engineer assignment dropdowns.
cmms_section_mst	294	SM_ID	0	Use for user display, sections/divisions, engineer assignment dropdowns.
departments	1	department_id	0	Use for user display, sections/divisions, engineer

				assignment dropdowns.
sections	2	section_id	2	Use for user display, sections/divisions, engineer assignment dropdowns.

7. Relationship Intelligence

HOW TO READ THIS

Foreign keys are the safety rails for service-layer transactions. For Phase 6 Slice 2, inspect every edge touching cmms_jobrequest_mst, cmms_jobcard_mst, cmms_equip_mst, users, employees, audit_log, and status-history tables before writing approve/assign SQL.

Core-module FK edges

Source table	Source cols	Target table	Target cols	Constraint
audit_log_changes	audit_id	audit_log	audit_id	fk_alc_audit
cmms_desig_hist	EMM_ID	cmms_emp_mst	EMM_ID	FK_CMMS_DESIG_HIST_CMMS_EMP_MST
cmms_division_hist	EQD_EQM_TYPE EQD_EQM_ID	cmms_equip_mst	EQM_TYPE EQM_ID	FK_CMMS_DIVISION_HIST_CMMS_EQIP_MST
cmms_emp_mst	EMM_ROLE	legacy_role_mst	ROLE_ID	FK_CMMS_EMP_MST_CMMS_ROLE_MST
cmms_emp_mst	EMM_DEPT	cmms_section_mst	SM_ID	FK_CMMS_EMP_MST_CMMS_SECTION_MST
cmms_equipinst_identification	EMD_EQIP_TYPE EQM_ID	cmms_equip_mst	EQM_TYPE EQM_ID	FK_CMMS_EQIPINST_IDENTIFICATION_CMMS_EQIP_MST
cmms_equip_detail_spec	EDS_EQIP_TYPE EDS_EQIP_ID	cmms_equip_mst	EQM_TYPE EQM_ID	FK_CMMS_EQIP_DETAIL_SPEC_CMMS_EQIP_MST
cmms_equip_mst	EQM_PCHKLSTNO	cmms_checklist_mst	CHKL_ID	FK_CMMS_EQIP_MST_CMMS_CHECKLIST_MST
cmms_equip_mst	EQM_CALCHKLSTNO	cmms_checklist_mst	CHKL_ID	FK_CMMS_EQIP_MST_CMMS_CHECKLIST_MST1
cmms_equip_mst	EQM_MFRID	cmms_cont_mst	CMM_CONT_ID	FK_CMMS_EQIP_MST_CMMS_CONT_MST
cmms_equip_mst	EQM_INST_TYPE	cmms_product_mst	PROD_ID	FK_CMMS_EQIP_MST_CMMS_PRODUCT_MST
cmms_equip_mst	EQM_DIVID	cmms_section_mst	SM_ID	FK_CMMS_EQIP_MST_CMMS_SECTION_MST
cmms_equip_mst	EQM_SECTION_ID	sections	section_id	fk_equip_section_new
cmms_equip_tec_spec	EMD_EQIP_TYPE EQM_ID	cmms_equip_mst	EQM_TYPE EQM_ID	FK_CMMS_EQIP_TEC_SPEC_CMMS_EQIP_MST
cmms_ins_accuracy_info	EIA_TYPE EIA_INS_ID	cmms_equip_mst	EQM_TYPE EQM_ID	FK_CMMS_INS_ACCURACY_INFO_CMMS_EQIP_MST
cmms_inv_equip_dtl	IND_EQM_TYPE IVD_EQM_ID	cmms_equip_mst	EQM_TYPE EQM_ID	FK_CMMS_INV_EQIP_DTL_CMMS_EQIP_MST
cmms_inv_equip_dtl	IVD_PARTNO	cmms_inv_mst	INV_PARTNO	FK_CMMS_INV_EQIP_DTL_CMMS_INV_MST
cmms_jobcard_attendedby_dtl	JMA_USERID	cmms_emp_mst	EMM_ID	FK_CMMS_JOBCARD_ATTENDEDDBY_DTL_CMMS_EMP_MST
cmms_jobcard_attendedby_dtl	JMA_SECTIONJOBNO	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_ATTENDEDDBY_DTL_CMMS_JOBCARD_MST
cmms_jobcard_awaitinginfo	JobcardNumber	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_AWAITINGINFO_CMMS_JOBCARD_MST
cmms_jobcard_cal_adjustments_dtl	JCAD_JobCardNo	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_CAL_ADJUSTMENTS_DTL_CMMS_JOBCARD_MST
cmms_jobcard_cal_dtl	JCD_JobCardNo	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_CAL_DTL_CMMS_JOBCARD_MST
cmms_jobcard_cal_observations	JobcardNumber	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_CAL_OBSERVATIONS_CMMS_JOBCARD_MST
cmms_jobcard_cal_observations	TaskID	cmms_task_mst	TSK_ID	FK_CMMS_JOBCARD_CAL_OBSERVATIONS_CMMS_TASK_MST
cmms_jobcard_contract_warranty_dtl	CWD_JobCardNo	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_CONTRACT_WARRANTY_DTL_CMMS_JOBCARD_MST
cmms_jobcard_eq_used	JEU_EquipType JEU_EquipId	cmms_equip_mst	EQM_TYPE EQM_ID	FK_CMMS_JOBCARD_EQ_USED_CMMS_EQIP_MST
cmms_jobcard_eq_used	JEU_JobCardNo	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_EQ_USED_CMMS_JOBCARD_MST
cmms_jobcard_faulty_category	FaultyCategory FaultyType	cmms_fault_mst	FM_FAULT_ID FM_TYPE	FK_CMMS_JOBCARD_FAULTY_CATEGORY_CMMS_FAULT_MST
cmms_jobcard_faulty_category	JobcardNumber	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_FAULTY_CATEGORY_CMMS_JOBCARD_MST
cmms_jobcard_faulty_section	FaultySection FaultyType	cmms_fault_mst	FM_FAULT_ID FM_TYPE	FK_CMMS_JOBCARD_FAULTY_SECTION_CMMS_FAULT_MST
cmms_jobcard_faulty_section	JobcardNumber	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_FAULTY_SECTION_CMMS_JOBCARD_MST
cmms_jobcard_inspection_info	JobcardNumber	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_INSPECTION_INFO_CMMS_JOBCARD_MST
cmms_jobcard_mst	JM_EQM_TYPE JM_EQM_ID	cmms_equip_mst	EQM_TYPE EQM_ID	FK_CMMS_JOBCARD_MST_CMMS_EQIP_MST
cmms_jobcard_repair_info	JobcardNumber	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_REPAIR_INFO_CMMS_JOBCARD_MST
cmms_jobcard_request_info	JRI_JobCardNo	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_REQUEST_INFO_CMMS_JOBCARD_MST
cmms_jobcard_request_item_dtl	JR_SECTIONJOBNO	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_REQUEST_ITEM_DTL_CMMS_JOBCARD_MST
cmms_jobcard_spares_equip	JobcardNumber	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_SPARES_EQIP_CMMS_JOBCARD_MST
cmms_jobcard_status_hist	JH_SectionJobNo	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBCARD_STATUS_HIST_CMMS_JOBCARD_MST
cmms_jobrequest_item_dtl	JR_JOBREQUESTNO	cmms_jobrequest_mst	JR_JOBREQUESTNO	FK_CMMS_JOBREQUEST_ITEM_DTL_CMMS_JOBREQUEST_MST
cmms_jobrequest_mst	JR_EQM_TYPE JR_EQM_ID	cmms_equip_mst	EQM_TYPE EQM_ID	FK_CMMS_JOBREQUEST_MST_CMMS_EQIP_MST
cmms_jobrequest_mst	JR_SECTIONJOB NO	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOBREQUEST_MST_CMMS_JOBCARD_MST
cmms_jobrequest_mst	JR_INST_TYPE	cmms_product_mst	PROD_ID	FK_CMMS_JOBREQUEST_MST_CMMS_PRODUCT_MST
cmms_jobrequest_mst	JR_DIVISION	cmms_section_mst	SM_ID	FK_CMMS_JOBREQUEST_MST_CMMS_SECTION_MST
cmms_jobrequest_project_dtl	JR_JOBREQUESTNO	cmms_jobrequest_mst	JR_JOBREQUESTNO	FK_CMMS_JOBREQUEST_PROJECTDTL_CMMS_JOBREQUEST_MST
cmms_jobrequest_project_dtl	JR_PROJECTID	cmms_proj_mst	PR_ID	FK_CMMS_JOBREQUEST_PROJECTDTL_CMMS_PROJ_MST
cmms_schedule_equip_dtl	SC_EQM_TYPE SC_EQM_ID	cmms_equip_mst	EQM_TYPE EQM_ID	FK_CMMS_SCHEDULE_EQIP_DTL_CMMS_EQIP_MST
cmms_schedule_equip_dtl	SC_PLAN_ID SC_TYPE	cmms_schedule_mst	SC_PLAN_ID SC_TYPE	FK_CMMS_SCHEDULE_EQIP_DTL_CMMS_SCHEDULE_MST
cmms_schedule_mst	SC_SM_ID	cmms_section_mst	SM_ID	FK_CMMS_SCHEDULE_MST_CMMS_SECTION_MST
equipment_status_history	transitioned by	cmms_emp_mst	EMM_ID	fk_esh_actor
equipment_status_history	eqm_type eqm_id	cmms_equip_mst	EQM_TYPE EQM_ID	fk_esh_equip
equipment_status_history	related job card	cmms_jobcard_mst	JM_SectionJobNo	fk_esh_jc
job_request_accessories	jr_no	cmms_jobrequest_mst	JR_JOBREQUESTNO	fk_jra_jr
job_request_status_history	transitioned by	cmms_emp_mst	EMM_ID	fk_jrsh_actor
job_request_status_history	jr_no	cmms_jobrequest_mst	JR_JOBREQUESTNO	fk_jrsh_jr
refresh_tokens	user_id	users	user_id	fk_rt_user
role_permissions	permission_id	permissions	permission_id	fk_rp_permission
role_permissions	role_id	roles	role_id	fk_rp_role
sections	department_id	departments	department_id	fk_sections_department
sections	head_employee_id	cmms_emp_mst	EMM_ID	fk_sections_head_emp
users	employee_id	cmms_emp_mst	EMM_ID	fk_users_employee
users	section_id	sections	section_id	fk_users_section

user_roles	role_id	roles	role_id	fk_ur_role
user_roles	user_id	users	user_id	fk_ur_user
legacy_jobcard_insp_maint_dtl	JMD_JobCardNo	cmms_jobcard_mst	JM_SectionJobNo	FK_CMMS_JOB_CARD_INSP_MAINT_DTL_CMMS_JOB_CARD_MST
legacy_section_user_mst	SM_ID	cmms_section_mst	SM_ID	FK_CMMS_SECTION_USER_MST_CMMS_SECTION_MST

8. State, Status, and ENUM Fields

STATE MACHINE WARNING

Every status/state/ENUM field is a potential state-machine boundary. For Phase 6, status transitions must go through a choke-point service/state-machine, not scattered UPDATE queries.

Table	Column	Type	Default	Module
cmms_checklist_hist	CHKL_STATE	tinyint(1)		Master Data / Lookup
cmms_checklist_mst	CHKL_STATE	tinyint(1)		Master Data / Lookup
cmms_cont_mst	CMM_CONT_TYPE	enum('MFR','VENDOR','BOTH','OEM')	'BOTH'	Vendor / Manufacturer
cmms_cont_mst	CMM_CONT_STATUS	varchar(100)	NULL	Vendor / Manufacturer
cmms_cont_mst	CMM_CONT_STATUS_FLAG	tinyint(1)	1	Vendor / Manufacturer
cmms_designation_mst	DG_STATE	tinyint(1)		Organization / Employee
cmms_device_spare_mst	DS_STATE	tinyint(1)		Master Data / Lookup
cmms_division_hist	EQD_STATUS	varchar(50)		Organization / Employee
cmms_division_hist	EQD_STATUS_DATE	datetime(6)		Organization / Employee
cmms_emp_mst	EMM_STATE	varchar(100)	NULL	Organization / Employee
cmms equip_mst	EQM_DIV_STATUS	varchar(50)	NULL	Equipment / Instruments
cmms equip_mst	EQM_STATUS_UPD_DATE	datetime(6)	NULL	Equipment / Instruments
cmms equip_mst	EQM_MVP_STATUS	enum('PENDING_VERIFICATION','ACTIVE','UNDER_CALIBRATION','UNDER_REPAIR','OUT_OF_TOLERANCE','QUARANTINED','CONDEMNED','RETIRED')	'PENDING_VERIFICATION'	Equipment / Instruments
cmms equip_mst	EQM_MVP_STATUS_AT	datetime(6)	NULL	Equipment / Instruments
cmms equip_mst_hist	EQM_DIV_STATUS	varchar(50)	NULL	Equipment / Instruments
cmms equip_mst_hist	EQM_STATUS_UPD_DATE	datetime(6)	NULL	Equipment / Instruments
cmms_inv_mst	INV_STATE	tinyint(1)		Procurement / AMC / Inventory
cmms_jobcard_awaitinginfo	RepairStatus	varchar(50)	NULL	Job Cards
cmms_jobcard_cal_dtl	JCD_CALStatus	varchar(50)	NULL	Job Cards
cmms_jobcard_cal_dtl	JCD_EQStatusAsRecd	varchar(50)	NULL	Job Cards
cmms_jobcard_cal_observations	Status	varchar(15)	NULL	Job Cards
cmms_jobcard_insp_section_info	EqRecdStatus	tinyint(3) UNSIGNED	NULL	Job Cards
cmms_jobcard_insp_section_info	FinalStatus	tinyint(3) UNSIGNED	NULL	Job Cards
cmms_jobcard_insp_section_info	FinalStatusDate	datetime(6)	NULL	Job Cards
cmms_jobcard_mst	JM_JobStatus	char(2)		Job Cards

cmms_jobcard_mst	JM_MVP_STATU S	enum('ASSIGNED','IN_PROGRESS','COMPLETED','VERIFIED_CLOSED','REOPENED')	'ASSIGNED'	Job Cards
cmms_jobcard_mst history	JM_JobStatus	char(2)		Job Cards
cmms_jobcard_repa ir_info	PrSystemStatus	tinyint(1)	NULL	Job Cards
cmms_jobcard_repa ir_info	PresentSystemSt atus	int(11)	NULL	Job Cards
cmms_jobcard_requ est_info	JRI_Working_Sta tus	varchar(20)	NULL	Job Cards
cmms_jobcard_stat us_hist	JH_JobStatus	varchar(2)		Job Cards
cmms_jobcard_stat us_hist	JH_StatusUpdate dOn	datetime(6)		Job Cards
cmms_jobcard_stat us_hist	JH_StatusUpdate dBy	varchar(7)		Job Cards
cmms_jobrequest_ mst	JR_WORKING_S TATUS	varchar(20)	NULL	Job Reques ts
cmms_jobrequest_ mst	JR_MVP_STATU S	enum('DRAFT','SUBMITTED','ASSIGNED','IN_PROGRESS','COMPLETED','VERIFIED_CLOSED','REJECTED','REOP ENED')	'DRAFT'	Job Reques ts
cmms_jobrequest_ mst	JR_MVP_STATU S_AT	datetime(6)	NULL	Job Reques ts
cmms_jobrequest_ mst	JR_PRIORITY	enum('LOW','NORMAL','HIGH','URGENT')	'NORMAL'	Job Reques ts
cmms_jobrequest_ mst	JR_JOB_CATEG ORY	enum('TME','FPE')	NULL	Job Reques ts
cmms_jobrequest_ mst	JR_JOB_TYPE	enum('CALIBRATION','REPAIR','REGISTRATION')	NULL	Job Reques ts
cmms_lineitem_mst	LITM_STATE	tinyint(1)		Master Data / Lookup
cmms_product_mst	PROD_STATE	tinyint(1)	0	Master Data / Lookup
cmms_proj_mst	PR_STATE	tinyint(1)		Master Data / Lookup
cmms_section_mst	SM_STATE	tinyint(1)		Organiz ation / Employ ee
cmms_task_mst	TSK_STATE	tinyint(1)		Master Data / Lookup
equipment_status_h istory	from_status	varchar(30)	NULL	Equipm ent / Instrum ents
equipment_status_h istory	to_status	varchar(30)		Equipm ent / Instrum ents
export_audit	export_type	enum('JOB_CARD_PDF','CAL_CERT_PDF','JOB_REQUEST_PDF','EXCEL_EQUIPMENT','EXCEL_JOB_CARDS')		Audit / Security
job_request_status_ history	from_status	varchar(30)	NULL	Job Reques ts
job_request_status_ history	to_status	varchar(30)		Job Reques ts
login_audit	outcome	enum('SUCCESS','FAILED_BAD_PASSWORD','FAILED_USER_LOCKED','FAILED_USER_INACTIVE','FAILED_NOT FOUND','FAILED_INVALID_FORMAT','LOGOUT','TOKEN_REFRESH')		Auth & RBAC
refresh_tokens	revoked_reason	enum('LOGOUT','ROTATED','ADMIN_REVOKE','PASSWORD_CHANGE','EXPIRY_CLEANUP')	NULL	Auth & RBAC
sections	equipment_categ ory	enum('TME','FPE')		Organiz ation / Employ ee
_legacy_cf001	CHKL_STATE	tinyint(1)		Legacy / Isolated
_legacy_cf003	CHKL_STATE	tinyint(1)		Legacy / Isolated
_legacy_userrole_m st	USER_STATE	tinyint(1)		Legacy / Isolated

9. Risk Audit - What Must Not Be Ignored

Risk count summary

Risk level	Count
HIGH	10
LOW	103
MEDIUM	177

High and medium risk observations

Table	Level	Type	Observation	Recommendation
audit_log	MEDIUM	POSSIBLE_MISSING_FK	Contains identifier-like columns but no FK parsed: audit_id, actor_employee_id, entity_id, request_id	Validate relationships before using joins or cascading lifecycle operations.
audit_log	MEDIUM	DATE_AS_TEXT	Column action appears temporal but is varchar(60).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_amc_mst	MEDIUM	DATE_AS_TEXT	Column CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_amc_mst	MEDIUM	DATE_AS_TEXT	Column UPDATED_ON appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_checklist_hist	HIGH	NO_PRIMARY_KEY	No primary key found in parsed DDL/ALTER statements.	Do not build write APIs on this table until a stable key strategy is confirmed.
cmms_checklist_hist	MEDIUM	DATE_AS_TEXT	Column CHKL_UPDATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_checklist_mst	MEDIUM	DATE_AS_TEXT	Column CHKL_CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_checklist_mst	MEDIUM	DATE_AS_TEXT	Column CHKL_UPDATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_checklist_tasks_hist	HIGH	NO_PRIMARY_KEY	No primary key found in parsed DDL/ALTER statements.	Do not build write APIs on this table until a stable key strategy is confirmed.
cmms_checklist_tasks_hist	MEDIUM	POSSIBLE_MISSING_FK	Contains identifier-like columns but no FK parsed: CLTSK_ID, CLTSK_TASKID	Validate relationships before using joins or cascading lifecycle operations.
cmms_cont_mst	MEDIUM	POSSIBLE_MISSING_FK	Contains identifier-like columns but no FK parsed: CMM_CONT_ID, CMM_CONT_NABL_CERT_NO	Validate relationships before using joins or cascading lifecycle operations.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_NAME appears temporal but is varchar(200).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_CONTACT_PERSON appears temporal but is varchar(150).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_EMAIL appears temporal but is varchar(150).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_PHONE appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_MOBILE appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_ADDRESS appears temporal but is varchar(500).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_CITY appears temporal but is varchar(100).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_STATE appears temporal but is varchar(100).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_COUNTRY appears temporal but is varchar(100).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_ZIP appears temporal but is varchar(20).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_WEBSITE appears temporal but is varchar(255).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_GSTIN appears temporal but is varchar(20).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_PAN appears temporal but is varchar(20).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_NABL_CERT_NO appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_REMARKS appears temporal but is varchar(1000).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_CREATED_BY appears temporal but is varchar(20).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_cont_mst	MEDIUM	DATE_AS_TEXT	Column CMM_CONT_UPDATED_BY appears temporal but is varchar(20).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_designation_mst	MEDIUM	DATE_AS_TEXT	Column DG_CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_designation_mst	MEDIUM	DATE_AS_TEXT	Column DG_UPDATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.

cmms_desig_hist	MEDIUM	DATE_AS_TEXT	Column EMM_DESIGNATION appears temporal but is varchar(200).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_device_spares_mst	MEDIUM	DATE_AS_TEXT	Column DS_CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_device_spares_mst	MEDIUM	DATE_AS_TEXT	Column DS_UPDATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_division_hist	HIGH	NO_PRIMARY_KEY	No primary key found in parsed DDL/ALTER statements.	Do not build write APIs on this table until a stable key strategy is confirmed.
cmms_division_hist	MEDIUM	DATE_AS_TEXT	Column EQD_STATUS appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_documentno_mst	MEDIUM	POSSIBLE_MISSING_FK	Contains identifier-like columns but no FK parsed: CurrentNo, MaximumNo, SequenceNo	Validate relationships before using joins or cascading lifecycle operations.
cmms_documentno_mst	MEDIUM	DATE_AS_TEXT	Column Description appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_emp_mst	MEDIUM	DATE_AS_TEXT	Column EMM_DESIGNATION appears temporal but is varchar(200).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_emp_mst	MEDIUM	DATE_AS_TEXT	Column EMM_STATE appears temporal but is varchar(100).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_emp_mst	MEDIUM	DATE_AS_TEXT	Column EMM_CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_emp_mst	MEDIUM	DATE_AS_TEXT	Column EMM_UPDATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst	MEDIUM	DATE_AS_TEXT	Column EQM_OPTIONNDESC appears temporal but is varchar(250).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst	MEDIUM	DATE_AS_TEXT	Column EQM_PONO appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst	MEDIUM	DATE_AS_TEXT	Column EQM_DIV_STATUS appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst	MEDIUM	DATE_AS_TEXT	Column EQM_CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst	MEDIUM	DATE_AS_TEXT	Column EQM_UPDATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst	MEDIUM	DATE_AS_TEXT	Column EQM_CAL_AT appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst_hist	MEDIUM	POSSIBLE_MISSING_FK	Contains identifier-like columns but no FK parsed: EQM_HIST_ID, EQM_ID, EQM_DIVID, EQM_PMCHKLSTNO, EQM_CALCHKLSTNO, EQM_TMPLTID, EQM_MFRID, EQM_VENID	Validate relationships before using joins or cascading lifecycle operations.
cmms_eqip_mst_hist	MEDIUM	DATE_AS_TEXT	Column EQM_OPTIONNDESC appears temporal but is varchar(250).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst_hist	MEDIUM	DATE_AS_TEXT	Column EQM_PONO appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst_hist	MEDIUM	DATE_AS_TEXT	Column EQM_DIV_STATUS appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst_hist	MEDIUM	DATE_AS_TEXT	Column EQM_CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst_hist	MEDIUM	DATE_AS_TEXT	Column EQM_UPDATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_eqip_mst_hist	MEDIUM	DATE_AS_TEXT	Column EQM_CAL_AT appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_fault_mst	MEDIUM	DATE_AS_TEXT	Column FM_CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_fault_mst	MEDIUM	DATE_AS_TEXT	Column FM_UPDATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_inv_mst	MEDIUM	DATE_AS_TEXT	Column INV_CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_inv_mst	MEDIUM	DATE_AS_TEXT	Column INV_UPDATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_jobcard_attendedby_dtl	MEDIUM	DATE_AS_TEXT	Column JMA_SECTIONJOBNO appears temporal but is varchar(9).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_jobcard_awaitinginfo	MEDIUM	DATE_AS_TEXT	Column RepairStatus appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_jobcard_awaitinginfo	MEDIUM	DATE_AS_TEXT	Column AttendedBy appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.
cmms_jobcard_cal_dtl	MEDIUM	DATE_AS_TEXT	Column JCD_CALStatus appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR_TO_DATE/validated parsing.

cmms_jobcard_cal_dtl	MEDIUM	DATE_AS_TEXT	Column JCD_ReasonForNoCAL appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_cal_dtl	MEDIUM	DATE_AS_TEXT	Column JCD_EQStatusAsRecd appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_cal_observations	MEDIUM	DATE_AS_TEXT	Column Status appears temporal but is varchar(15).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_contract_warranty_dtl	MEDIUM	DATE_AS_TEXT	Column CWD_GatePassNo appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_contract_warranty_dtl	MEDIUM	DATE_AS_TEXT	Column CWD_GatePassIssuedBy appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_inspection_info	MEDIUM	DATE_AS_TEXT	Column EqNotRecdReason appears temporal but is varchar(500).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_inspection_info	MEDIUM	DATE_AS_TEXT	Column InspectionResult appears temporal but is varchar(20).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_inspection_info	MEDIUM	DATE_AS_TEXT	Column RejectionReasons appears temporal but is varchar(200).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_inspection_info	MEDIUM	DATE_AS_TEXT	Column FaultDescription appears temporal but is varchar(200).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_inspection_info	MEDIUM	DATE_AS_TEXT	Column ActionTakenBySupplier appears temporal but is varchar(100).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst	MEDIUM	DATE_AS_TEXT	Column JM_SectionJobNo appears temporal but is varchar(9).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst	MEDIUM	DATE_AS_TEXT	Column JM_JobStatus appears temporal but is char(2).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst	MEDIUM	DATE_AS_TEXT	Column JM_AttendedBy appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst	MEDIUM	DATE_AS_TEXT	Column JM_CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst	MEDIUM	DATE_AS_TEXT	Column JM_UPDATED_BY appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst	MEDIUM	DATE_AS_TEXT	Column JM_REOPENED_REASON appears temporal but is varchar(500).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst_history	MEDIUM	POSSIBLE_MISSING_FK	Contains identifier-like columns but no FK parsed: HistoryId, JM_JobCardNO, JM_EQM_ID, JM_SectionJobNo, JM_PLANID	Validate relationships before using joins or cascading lifecycle operations.
cmms_jobcard_mst_history	MEDIUM	DATE_AS_TEXT	Column JM_SectionJobNo appears temporal but is varchar(9).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst_history	MEDIUM	DATE_AS_TEXT	Column JM_JobStatus appears temporal but is char(2).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst_history	MEDIUM	DATE_AS_TEXT	Column JM_AttendedBy appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst_history	MEDIUM	DATE_AS_TEXT	Column JM_CREATED_BY appears temporal but is varchar(7).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst_history	MEDIUM	DATE_AS_TEXT	Column JM_UPDATED_BY appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_mst_history	MEDIUM	DATE_AS_TEXT	Column HU_UPDATED_BY appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_repair_info	MEDIUM	DATE_AS_TEXT	Column ReasonsForNotRepaired appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_repair_info	MEDIUM	DATE_AS_TEXT	Column FaultDescription appears temporal but is varchar(550).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_repair_info	MEDIUM	DATE_AS_TEXT	Column ActionTaken appears temporal but is varchar(550).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_request_info	MEDIUM	DATE_AS_TEXT	Column JRI_Designation appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.
cmms_jobcard_request_info	MEDIUM	DATE_AS_TEXT	Column JRI_Division appears temporal but is varchar(50).	Normalize in API layer; avoid direct date arithmetic without STR TO DATE/validated parsing.

10. Locked Engineering Rules for Next Build

Rule	Why
No destructive schema changes	The database contains live/historical value. Additive migration only unless explicitly approved.
No direct status UPDATE	Job Request and Job Card statuses must pass through state-machine choke-points.

Audit is part of the transaction	A state change without audit is not acceptable in government/precision engineering context.
Row-level scope is server-side	Normal users and engineers must never rely on frontend filtering for visibility.
Use exact legacy column names in repositories	The dump is the source of truth; no invented columns.
legacy* remains read-only	Legacy tables are preserved for audit/history and should not become new write surfaces.
Run review SQL after import	Parsed dump evidence must be verified against physical imported DB metadata.

11. Full Table Inventory Appendix

Table	Module	Use	Row s	Col s	PK	Id x	F K
audit_log	Audit / Security	ACTIVE_CORE_SECURITY	20	11	audit_id	5	0
audit_log_changes	Audit / Security	ACTIVE_CORE_SECURITY	0	5	change_id	2	1
cmms_amc_mst	Procurement / AMC / Inventory	FUTURE_PHASE_CORE	0	11	AMC_ID	3	2
cmms_checklist_hist	Master Data / Lookup	ACTIVE_SUPPORT_MAST ER	811	7	MISSING	0	0
cmms_checklist_mst	Master Data / Lookup	ACTIVE_SUPPORT_MAST ER	928	9	CHKL_ID	2	1
cmms_checklist_tasks	Master Data / Lookup	ACTIVE_SUPPORT_MAST ER	7,536	2	CLTSK_ID;CLTSK_TASKID	2	2
cmms_checklist_tasks_hist	Master Data / Lookup	ACTIVE_SUPPORT_MAST ER	8,450	4	MISSING	0	0
cmms_cont_mst	Vendor / Manufacturer	ACTIVE_SUPPORT_MAST ER	540	23	CMM_CONT_ID	5	0
cmms_designation_mst	Organizatio n / Employee	ACTIVE_SUPPORT_MAST ER	40	8	DG_ID	1	0
cmms_desig_hist	Organizatio n / Employee	ACTIVE_SUPPORT_MAST ER	132	3	EMM_ID;EMM_DESIGNATION;EMM_DATE	1	1
cmms_device_spares_mst	Master Data / Lookup	ACTIVE_SUPPORT_MAST ER	67	8	DS_ID	1	0
cmms_division_hist	Organizatio n / Employee	ACTIVE_SUPPORT_MAST ER	3,676	7	MISSING	1	1
cmms_documentno_mst	Other / Cross-cutting	REVIEW_REQUIRED	151	6	DocumentType;DocumentYear	1	0
cmms_emp_mst	Organizatio n / Employee	ACTIVE_SUPPORT_MAST ER	60	30	EMM_ID	4	2
cmms_eqipinst_identification	Equipment / Instruments	ACTIVE_MVP_CORE	2,286	10	EMD_EQIP_TYPE;EQM_ID;EII_ID	1	1
cmms_eqip_detail_spec	Equipment / Instruments	ACTIVE_MVP_CORE	3	4	EDS_EQIP_TYPE;EDS_EQIP_ID;EDS_FILENAME	1	1
cmms_eqip_mst	Equipment / Instruments	ACTIVE_MVP_CORE	5,705	52	EQM_TYPE;EQM_ID	10	6
cmms_eqip_mst_hist	Equipment / Instruments	ACTIVE_MVP_CORE	519	49	EQM_HIST_ID	1	0
cmms_eqip_tec_spec	Equipment / Instruments	ACTIVE_MVP_CORE	0	4	EMD_EQIP_TYPE;EQM_ID;EMD_SPEC_NAME	1	1
cmms_fault_mst	Job Cards	ACTIVE MVP CORE	30	8	FM_FAULT_ID;FM_TYPE	1	0
cmms_ins_accuracy_info	Equipment / Instruments	ACTIVE_MVP_CORE	1,501	5	EIA_INS_ID;EIA_TYPE;EIA_ACCURACY	2	1
cmms_inv_eqip_dtl	Equipment / Instruments	ACTIVE_MVP_CORE	24	3	IVD_PARTNO;IVD_EQM_ID;IND_EQM_TYPE	2	2
cmms_inv_mst	Procurement / AMC / Inventory	FUTURE_PHASE_CORE	42	29	INV_PARTNO	3	2
cmms_jobcard_attendedby_dtl	Job Cards	ACTIVE_MVP_CORE	27,890	4	JMA_SECTION;JOBNO;JMA_USERID;JMA_ISAWAITING	2	2
cmms_jobcard_awaitinginfo	Job Cards	ACTIVE MVP CORE	7,261	19	JobcardNumber	1	1
cmms_jobcard_cal_adjustments_dtl	Job Cards	ACTIVE_MVP_CORE	1,831	6	JCAD_JobCardNo;JCAD_Parameter;JCAD_TestValue	1	1
cmms_jobcard_cal_dtl	Job Cards	ACTIVE MVP CORE	9,065	21	JCD_JobCardNo	1	1
cmms_jobcard_cal_observations	Job Cards	ACTIVE_MVP_CORE	77,171	4	JobcardNumber;TaskID	2	2
cmms_jobcard_contract_warranty_dtl	Job Cards	ACTIVE_MVP_CORE	17,225	17	CWD_JobCardNo	1	1
cmms_jobcard_eq_used	Job Cards	ACTIVE_MVP_CORE	38,316	4	JEU_JobCardNo;JEU_EquipType;JEU_EquipId	2	2
cmms_jobcard_faulty_category	Job Cards	ACTIVE MVP CORE	8,605	3	JobcardNumber;FaultyType;FaultyCategory	2	2
cmms_jobcard_faulty_section	Job Cards	ACTIVE MVP CORE	8,131	3	JobcardNumber;FaultyType;FaultySection	2	2
cmms_jobcard_inspection_info	Job Cards	ACTIVE MVP CORE	2,214	21	JobcardNumber	1	1
cmms_jobcard_mst	Job Cards	ACTIVE_MVP_CORE	19,432	38	JM_SectionJobNo	6	1
cmms_jobcard_mst_history	Job Cards	ACTIVE_MVP_CORE	22,143	37	HistoryId	1	0
cmms_jobcard_repair_info	Job Cards	ACTIVE MVP CORE	8,118	18	JobcardNumber	1	1
cmms_jobcard_request_info	Job Cards	ACTIVE_MVP_CORE	19,432	27	JRI_JobCardNo	1	1
cmms_jobcard_request_item_dtl	Job Cards	ACTIVE_MVP_CORE	11,064	10	JR_SECTION;JOBNO;JR_ITEM_ID	1	1
cmms_jobcard_spares equip	Job Cards	ACTIVE MVP CORE	2,804	9	JobcardNumber;Sr_No	2	1

cmms_jobcard_status_hist	Job Cards	ACTIVE_MVP_CORE	22,214	5	MISSING	1	1
cmms_jobrequest_item_dtl	Job Requests	ACTIVE_MVP_CORE	7,786	11	JR_JOBREQUESTNO;JR_ITEM_ID	1	1
cmms_jobrequest_mst	Job Requests	ACTIVE_MVP_CORE	21,490	52	JR_JOBREQUESTNO	13	4
cmms_jobrequest_project_dtl	Job Requests	ACTIVE_MVP_CORE	19,624	2	JR_JOBREQUESTNO;JR_PROJECTID	2	2
cmms_lineitem_mst	Master Data / Lookup	ACTIVE_SUPPORT_MAST ER	24	9	LITM_ID	1	0
cmms_parameter_master	Master Data / Lookup	ACTIVE_SUPPORT_MAST ER	367	10	CategoryID;SrID	2	0
cmms_po_item_dtl	Procurement / AMC / Inventory	FUTURE_PHASE_CORE	303	9	MISSING	2	2
cmms_po_mst	Procurement / AMC / Inventory	FUTURE_PHASE_CORE	115	16	PO_ID	1	0
cmms_po_payment_dtl	Procurement / AMC / Inventory	FUTURE_PHASE_CORE	187	5	MISSING	1	1
cmms_po_receive_dtl	Procurement / AMC / Inventory	FUTURE_PHASE_CORE	42	3	MISSING	1	1
cmms_product_mst	Master Data / Lookup	ACTIVE_SUPPORT_MAST ER	32	11	PROD_ID	1	0
cmms_proj_mst	Master Data / Lookup	ACTIVE_SUPPORT_MAST ER	182	7	PR_ID	1	0
cmms_pur_dtl	Procurement / AMC / Inventory	FUTURE_PHASE_CORE	0	13	PUD_NO;PUD_PARTNO	1	1
cmms_pur_mst	Procurement / AMC / Inventory	FUTURE_PHASE_CORE	0	22	PUM_NO	1	0
cmms_schedule_eqip_dtl	Equipment / Instruments	ACTIVE_MVP_CORE	316	5	SC_PLAN_ID;SC_TYPE;SC_EQM_ID;SC_EQM_TYPE;SC_SCHEDULE_DATE	2	2
cmms_schedule_mst	Scheduling / PM-CAL	FUTURE_PHASE_CORE	6	8	SC_PLAN_ID;SC_TYPE	2	1
cmms_section_mst	Organization / Employee	ACTIVE_SUPPORT_MAST ER	294	14	SM_ID	1	0
cmms_supplier_mfr	Vendor / Manufacturer	ACTIVE_SUPPORT_MAST ER	159	3	MISSING	3	3
cmms_task_mst	Master Data / Lookup	ACTIVE_SUPPORT_MAST ER	1,489	11	TSK_ID	1	0
cmms_vendor_doc_spec	Vendor / Manufacturer	ACTIVE_SUPPORT_MAST ER	0	4	MISSING	0	0
departments	Organization / Employee	ACTIVE_SUPPORT_MAST ER	1	9	department_id	2	0
equipment_status_history	Equipment / Instruments	ACTIVE_MVP_CORE	0	9	history_id	5	3
export_audit	Audit / Security	ACTIVE_CORE_SECURITY	0	7	export_id	4	0
job_request_accessories	Job Requests	ACTIVE_MVP_CORE	3	7	acc_id	2	1
job_request_status_history	Job Requests	ACTIVE_MVP_CORE	8	7	history_id	4	2
login_audit	Auth & RBAC	ACTIVE_CORE_SECURITY	57	7	audit_id	4	0
permissions	Auth & RBAC	ACTIVE_CORE_SECURITY	40	7	permission_id	3	0
refresh_tokens	Auth & RBAC	ACTIVE_CORE_SECURITY	43	9	token_id	4	1
roles	Auth & RBAC	ACTIVE_CORE_SECURITY	5	6	role_id	2	0
role_permissions	Auth & RBAC	ACTIVE_CORE_SECURITY	122	4	role_id;permission_id	2	2
schema_migrations	Other / Cross-cutting	REVIEW_REQUIRED	15	6	migration_id	1	0
sections	Organization / Employee	ACTIVE_SUPPORT_MAST ER	2	12	section_id	6	2
sysdiagrams	Other / Cross-cutting	REVIEW_REQUIRED	0	5	diagram_id	2	0
tmpcaldtl	Other / Cross-cutting	REVIEW_REQUIRED	732	3	MISSING	0	0
users	Auth & RBAC	ACTIVE_CORE_SECURITY	60	14	user_id	5	2
user_roles	Auth & RBAC	ACTIVE_CORE_SECURITY	60	4	user_id	2	2
_legacy_accessright_mst	Legacy / Isolated	LEGACY_ISOLATED	3,221	11	ACC_MOD_ID;ACC_ROLE	2	2
_legacy_cal_jobcard_feedback_spec	Legacy / Isolated	LEGACY_ISOLATED	0	4	MISSING	0	0
_legacy_cf001	Legacy / Isolated	LEGACY_ISOLATED	6	7	MISSING	0	0
_legacy_cf002	Legacy / Isolated	LEGACY_ISOLATED	553	4	MISSING	0	0
_legacy_cf003	Legacy / Isolated	LEGACY_ISOLATED	570	9	MISSING	0	0
_legacy_cf004	Legacy / Isolated	LEGACY_ISOLATED	3,449	2	MISSING	0	0
_legacy_chklistvendor	Legacy /	LEGACY_ISOLATED	238	3	MISSING	0	0

	Isolated						
_legacy_jobcard_insp_maint_dtl	Legacy / Isolated	LEGACY_ISOLATED	0	6	JMD_JobCardNo	1	1
_legacy_module_mst	Legacy / Isolated	LEGACY_ISOLATED	163	13	MOD_ID	2	1
_legacy_parameter_master_bkp	Legacy / Isolated	LEGACY_ISOLATED	4	4	MISSING	0	0
_legacy_parameter_master_incharge	Legacy / Isolated	LEGACY_ISOLATED	9	4	MISSING	0	0
_legacy_parameter_master_jun2016	Legacy / Isolated	LEGACY_ISOLATED	233	4	MISSING	0	0
_legacy_role_mst	Legacy / Isolated	LEGACY_ISOLATED	23	7	ROLE_ID	1	0
_legacy_section_user_mst	Legacy / Isolated	LEGACY_ISOLATED	294	4	SM_ID;SM_USER_ID	2	2
_legacy_userrole_mst	Legacy / Isolated	LEGACY_ISOLATED	565	9	USER_ID;USER_DIVISION_ID	1	0

12. Immediate Next Action Plan

Step	Action	Owner
1	Import dump into local/staging DB and run sql/01_senior_review_queries.sql.	DS / Backend
2	Open CSV 04 and 05 before writing Phase 6 Slice 2 joins.	Backend
3	Resolve HIGH risk rows in 07_table_risk_audit.csv, especially missing PK tables that may become write paths.	Backend + DB
4	For Slice 2, implement JR detail + approve/reject + assign + auto-create JC as one transaction.	Backend
5	Create frontend detail and LIC action panels only after backend service contract is stable.	Frontend
6	Update locked documentation after browser smoke tests pass.	DS Documentation

FINAL LOCK

This report is the dump90tables evidence backbone. Use the CSV/SQL/JSON artifacts for exact implementation. Use this DOCX for senior-level understanding, review, and handover.